ID	Name	Email Address	Title of the presentation	Abstract	Please	Please specify your Department:	Please categorize the modelling technique	Simula	Optim	Accou	AI ar
1	Danial Esmaeilialiabadi	danial.esmaeili@ufz.de	The emerging threat of artificial intelligence on competition in liberalized electricity markets: A deep Q-network approach	According to Sustainable Development Goals (SDGs), societies should have access to affordable, reliable, and sustainable energy. Liberalized electricity markets have been established to provide affordable electricity for end-users through advertising competition. Although these new markets are designed to serve competition, there are recorded incidents where participants abused their market power and disrupted the competition through collusion. Unfortunately, modern autonomous pricing algorithms may further assist myopic players to discover collusive strategies with a minimum amount of sensitive information. Therefore, in this study, we investigate the impact of emerging learning algorithms on the bidding strategies of Power Generating Companies (GenCos) and compare their performance against game-theoretic expectations. A novel deep Q-network (DQN) model is developed, by which GenCos determine the bidding strategies to maximize average long-term payoffs in a day-ahead market. The presented DQN model assumes that GenCos have no information regarding the rivals' true generation costs and profits. To the best of the authors' knowledge, this is the first study that thoroughly investigates players' behavior utilizing a modern DQN model and compares its results with equilibria of the non-cooperative single-stage and infinitely-repeated games in the context of electricity markets. The outcomes articulate that GenCos equipped with advanced learning models may be able to collude unintentionally while trying to ameliorate long-term profits. Moreover, GenCos that employ the presented DQN model could discover and sustain more profitable (e.g., collusive) strategies vis-à-vis a conventional Q-learning method. Collusive strategies can lead to exorbitant electric bills for end-users, which is one of the influential factors in energy poverty. Thus, policymakers and market designers should be vigilant regarding the combined effect of information disclosure and autonomous pricing, as new models exploit information more effectively.	6	Bioenergy	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM); Simulation Model (e	Nein	Ja	Nein	Ja
2	Walther Zeug	walther.zeug@ufz.de	Holistic and Integrated Life Cycle Sustainability Assessment: Background, Methods and Results from Two Case Studies	To address the challenges of the current societal-ecological crisis, we developed an integrated sustainability framework and an implemented and validated innovative method of Holistic and Integrated Life Cycle Sustainability Assessment (HILCSA). HILCSA allows an integrative (ecological, economic, social in one method) and holistic (transdisciplinary and critical) sustainability assessment based on about 100 social, ecological and economic qualitative and quantitative indicators addressing 14 out of 17 SDGs, in order to analyze synergies, trade-offs and hotspots of production and consumption systems in the bioeconomy and beyond. This method is fully software implemented in openLCA and using the Ecoinvent/SoCa database. Through this quantitative and qualitative sustainability assessments we identify synergies and hot-spots of bioeconomy production systems production on a detailed and aggregated level. Common problems are the very hard planetary boundary of land and water availability limiting renewable resource and goods production, as well as maintained global socio-economic problems in supply chains when bioeconomy does not go in hand with a societal-ecological transformation. HILCSA can provide comprehensive information and decision basis for stakeholders such as politics, society, research and organizations. However, problems and potentials in method, software and indicator development still need to be discussed in future research.	6	Bioenergy	Accounting Model (e.g., LCA)				
3	Mariana Madruga de Brito	mariana.brito@ufz.de	Advancing hydrological extremes research through NLP	Much of the scientific findings, public discourse, and political framing related to climate change impacts and adaptation are conveyed through text. The growing volume of textual data, combined with advancements in natural language processing (NLP), has opened up previously unforeseen opportunities for investigating the social aspects of hydrological extremes. Our working group uses text mining on various text corpora to deepen our understanding of how these events impact society and how we adapt to them. For instance, by analyzing peer-reviewed literature and Wikipedia articles, we extract insights into the impacts of natural hazards. We also examine texts such as water board committee meeting minutes to uncover how water allocation decisions are made and identify influential stakeholders. Additionally, newspaper data allows us to assess public perception of the socioeconomic effects of floods. Our research extends to monitoring adaptation measures worldwide based on government reports and climate adaptation plans. In summary, integrating NLP into hydrological research offers a powerful tool for uncovering the complex social, political, and economic dynamics shaping risk.	6	Department of Urban and Environmental Sociology	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
4	Daniela Thrän	daniela.thraen@ufz.de	The Opening	will follow	6	Bioenergy	Accounting Model (e.g., LCA)				
5	Mohammad Sadr	mohammad.sadr@ufz.de	Optimizing Bioenergy with Carbon Capture and Storage (BECCS) for Climate Mitigation	Bioenergy with Carbon Capture and Storage (BECCS) is a promising negative emission technology (NET) for mitigating climate change. However, its large-scale deployment faces significant challenges, including techno-economic constraints and policy uncertainties. To address these challenges, we employ a spatially detailed optimization model to assess the potential of BECCS in a specific region (i.e., Germany). Our model incorporates key factors such as biomass availability, infrastructure costs, and carbon pricing to identify optimal BECCS deployment scenarios. By systematically analyzing various scenarios, we aim to quantify the maximum achievable carbon removal potential and identify the most cost-effective pathways. Our findings highlight the critical role of policy instruments, such as carbon pricing and subsidies, in incentivizing BECCS deployment. By optimizing the design and operation of BECCS systems, we can contribute to achieving ambitious climate targets and mitigating the impacts of climate change.	6	Bioenergy	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs)				
6	Matthias Jordan	matthias.jordan@ufz.de	BENOPTex: Modeling scenarios for the transformation of the German bioeconomy and energy system with a linear optimization model.	BENOPTex: Modeling scenarios for the transformation of the German bioeconomy and energy system with a linear optimization model. The analysis aims on identifying the optimal use of various biomass types within the future bioeconomy and energy system. The focus of the analysis is on sectoral details and on high quality biomass data (feedstocks, prices, conversion technologies). Consequently, the model has to represent a lot of interfaces/ synergies, which exist for biomass usage. Besides the consideration of the energy and material demand, the carbon flow is modeled. Regarding data, the model is connected to a SQL database for technological data, a json dump including data on negative emission technologies and reading a variety of Excel files. It relies on data from various scources, e.g. Öko-Institut, DBFZ and UFZ. It uses a python function developed at the Forschungszentrum Jülich to aggregate highly resolved temporal data.	6	Bioenergy	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs)				
7	Rohini Kumar	rohini.kumar@ufz.de	TBC	TBC	5	CHS	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM); Simulation Model (e.g., agent-based models, econometrics ar				
8	Ali Forootani	ali.forootani@ufz.de	Climate aware Deep Neural Network for Wind Power Forecast	WIND power forecasting using climate data is a crucial aspect of modern energy systems, enabling better integration of renewable energy into the grid. By leveraging climate datasets—such as wind speed, pressure, temperature, and atmospheric conditions—predictive models can estimate future wind energy output with greater accuracy. This is vital for managing the intermittency of wind power, optimizing energy dispatch, and ensuring grid stability. This study refines wind energy forecasting in Germany by integrating CMIP6 climate data with advanced machine learning-compatible techniques, including temporal resampling, spatial interpolation, and Min-Max normalization. These methods standardize large datasets for models like LSTM networks, optimizing data for sequential forecasting. We also explores deep learning models—MLPs , LSTMs for sequential dependencies, and transformer-enhanced LSTM models—to improve prediction accuracy for wind power forecasting.	1	Dep. Bioenergy, Helmholtz Center for Environmental Research-UFZ	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
9	Almudena García-García	almudena.garcia-garcia@ufz.de	Generation and evaluation of energy and water fluxes from the HOLAPS framework	Improving our understanding of the energy and water exchanges between the land surface and the lower atmosphere (i.e. land-atmosphere interactions), and how climate change may affect them, is crucial to predict changes in temperature and precipitation extremes. Observations of energy and water fluxes at the land surface are typically retrieved from the eddy covariance method, which presents limitations related to spatial and temporal gaps, and the non-closure of the energy and water balances. Meanwhile, soil moisture (SM) products derived from satellite data have been widely used at regional and global scales, but they are limited to capture only surface soil water content and variations. The combination of remote sensing (RS) data and modelling frameworks is called to be the solution to improve the spatial coverage and vertical resolution of land-atmosphere interactions data, ensuring the energy and water balance closure. Here, we explore the combination of remote sensing and meteorological data with a physical-based modeling framework, the High resOlution Land Atmosphere Parameters from Space (HOLAPS). We used HOLAPS to produce hourly consistent estimates of energy and water fluxes over Europe at 5 km resolution. HOLAPS and other satellite-based evapotranspiration and sensible heat flux products from the literature are evaluated against the water balance method and eddy covariance measurements. HOLAPS SM estimates together with other RS-modelling products are also evaluated against ground-based measurements at the surface and in the root zone. The evaluation of HOLAPS ET estimates show similar performance to the other products with Kling–Gupta efficiency (KGE) > -0.41 in comparison with eddy covariance measurements from FLUXNET in all seasons but in boreal winter. The simulation of H is more uncertain than for ET with KGE values ranging from -2.5 to 0.8 along the products and stations at monthly scales. HOLAPS reaches slightly better results than the rest of ET and H products at daily scales in summer (KGE > 0.3 for ET and KGE > 0.0 for H) and during hot conditions (KGE > 0.2 for ET and KGE >-0.2 for H), while the state-of-the-	5	Remote Sensing Department	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
10	Francisco José Cuesta-Valero	francisco-jose.cuesta-valero@ufz.de	Ground heat flux estimates from remote sensing products and machine learning models	Energy exchanges at the land surface are important to understand the long-term and short-term interactions between the lower atmosphere and the shallow subsurface. The energy exchanges between the land and the atmosphere are governed by the surface energy budget, that is, the partition of the net surface radiation absorbed by the land into latent heat flux (associated with evapotranspiration), sensible heat flux (associated with heat conduction at the land-air interface), and ground heat flux (associated with heat conduction through the subsurface). Therefore, knowing the evolution of ground heat flux is critical to close the surface energy balance and reduce the uncertainty in estimates of latent and sensible heat flux. Determining the long-term, global ground heat flux is also relevant for understanding and quantifying climate change: ground heat storage, estimated from changes in ground heat flux, is the second largest term of the Earth heat inventory only after the ocean. Despite the importance of ground heat flux at at different temporal scales, measurements are markedly scarce and unevenly distributed. Estimates of ground heat flux from subsurface temperature profiles have been used to mitigate this lack of direct measurements, but such estimates consist of decadal to centennial changes in heat flux. Furthermore, the global network of temperature profiles presents important geographical and temporal gaps, being unable to produce a global estimate of ground heat flux in recent times. Only recently estimates of global heat storage have been derived from gridded meteorological	5		AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM); Simulation Model (e.g., agent-based models, econometrics ar				
11	Sandra Gutjahr	sandra.gutjahr@ufz.de	Adding stochastic programming to Bioenergy optimization model	The work presents the integration of stochastic programming into the Bioenergy optimization model BENOPTex to analyse the interplay between VRE and dispatchable energy sources. Bioenergy with its inherent flexibility, can mitigate the negative impacts of the intermittency of renewables on a net-zero emissions energy system and stochastic programming captures uncertainty from past weather data in the energy system model. In deterministic models, decisions are based on single operational scenarios with a probability of realization set at 100%, while stochastic models define scenarios by considering the likelihood of different outcomes. Previous studies oftentimes introduced equiprobable scenarios, whereas in this study, the probability of occurrence for different possible scenarios is derived. Outcomes suggest that deterministic models provide an optimistic solution which does not account for the stochastic nature of VRE. Preliminary findings demonstrate that for large amount of solar and wind energy the overall system cost is at the minimum whereas during shortages, flexible bioenergy can support meeting electricity demand but at higher system cost.	6	Bioenergy	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs)				
12	Annegret Haase	annegret.haase@ufz.de	Social science thinking and big data and modeling	I am interested in this workshop from a social science perspective. I have no experience with working with big data and modelling so far but want to better understand how it can be used for working on social science questions. I will not have a presentation at the workshop but want to discuss this question with people who are experienced in working with big data and modelling. For the categorizations below, I just clicked on to be able to finish the registration.	6	SUSOZ	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
13	Taimur Khan	taimur.khan@ufz.de	DeepTrees: Deep-Learning based spatiotemporal tree inventorying from public orthoimages	Field-based methods for tree inventorying are time-consuming, labour-intensive, and often expensive. With the advent of remote sensing technologies and deep learning, it is now possible to leverage digital public orthoimages and artificial intelligence to automate and enhance the tree inventorying process [1]. The DeepTrees project aims to develop and implement deep learning models for tree crown segmentation, tree trait detection and tree species classification for the publicly available imagery from the Digital Orthoimages Program (DOP) at 20 cm scale in Germany. The resulting tree inventories and monitoring data can be used for various applications, including forest management, biodiversity monitoring, and ecological research. Segmented tree crowns can give a wealth of information about the individual trees. The significance of a tree's fitness is closely tied to both the structure and size of its crown [2]. These factors influence the tree's access to resources, its utilisation and occupation of space, the magnitude of its growth, as well as seed production and dispersal. In densely populated environments, the growth of crown size results in competition for space, leading to social differentiation, reduced growth of suppressed trees, mortality, and self-thinning [3]. Furthermore, the practical and economic relevance of crown structure and size should not be	1	Community Ecology	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
14	Luis Samaniego	luis.samaniego@ufz.de	not available now	Lorem Ipsum ...	5	CHS	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
15	Kabiru Nata'ala Muhammad	muhammad.kabiru@ufz.de	Unveiling the carbon fixation potential of marine Prokaryotes using MarMAGs, a novel user-friendly database with over 130 thousand metagenome assemble genomes	Our team designed the MarMAGs, a user-friendly database containing over 130 thousand marine metagenome-assembled genomes (MAGs). Users can access this data from coordinates in a map or by filtering it based on its metadata (e.g., marine biome, biogeographic features, and sequencing platform). Further, we showcased carbon fixation as a use case. We recovered metagenome-assembled genomes (MAGs) from 3405 marine metagenomes selected from MarineMetagenomeDB (https://web.app.ufz.de/marmdb/) and added 52162 MAGs from the OceanDNA dataset and 8558 marine MAGs from the GEM catalog. We evaluated genome quality, taxonomy, and assembly. After metadata harmonization, we identified the genes for prokaryotes' carbon fixation pathways that included the Reductive pentose phosphate cycle (CBB), Reverse Citrate cycle, 3-Hydroxypropionate bicycle (3HPB), Dicarboxylate-hydroxybutyrate cycle (DCHB), Hydroxypropionate Hydroxybutyrate cycle, and Reductive Acetyl-CoA pathway. Due to the complexity of reversible enzymatic reactions and the versatility of the involved enzymes, the Reductive Glycine pathway and reverse oxidative TCA cycle were not annotated in our analysis. We recovered 73,944 novel MAGs that, together with the other two MAG datasets, comprised the MarMAGs with 134,644 MAGs. The MarMAGs contain 31.2%(42047) high-quality and 68.78%(92617) medium-quality MAGs. Bacteria dominate the taxonomic composition of the MAGs at 89.89% (121046), and archaea at 10.11% (13618) spread across 135 phyla. Our analyses revealed 3.96% (5333) of the MAGs with genetic potential for at least one carbon fixation pathway, with CBB comprising 79.75% (4253), while 3HPB 0.02% (1) and DCHB 0.08% (4) have the least of them. Also, 28 MAGs affiliated to 5 phyla have the genetic potential for two pathways. Additionally, we are conducting	1	Department of Applied Microbial Ecology	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM); Optimization Model (e.g., linear programming, mixed-integer				
16	Jana Schor	jana.schor@ufz.de	Streamlining Data Access: An LLM Chatbot for Eco-Toxicological Knowledge Graphs	EcoToxFred, our domain-specific chatbot prototype, integrates retrieval-augmented generation (RAG) for large language models (LLMs) to facilitate human-like interactions with eco-toxicological knowledge stored in a graph database. In particular, we generated a knowledge graph from a huge integrated dataset of measured chemical exposure in European surface waters (chemical monitoring data) with hazard information for aquatic species. Inferred values like toxic units and summarized toxic units for sampling sites have been calculated for this data. The collected monitoring information covers a time frame of the last 25 years. We used a Neo4j graph database to store this knowledge. Within EcoToxFred, we developed a RAG-powered LLM chatbot prototype to allow non-computer scientists and stakeholders access to this domain-specific knowledge graph. Our approach is highly transferable to any research area where complex data relationships are essential. By utilizing structured graph databases, LLMs can retrieve interconnected insights from vast datasets, supporting in-depth analysis and decision-making without requiring users to have advanced technical skills. EcoToxFred exemplifies a scalable, flexible approach for enhancing data accessibility and cross-disciplinary knowledge transfer, demonstrating the potential of LLM-knowledge graph systems to streamline knowledge extraction and broaden collaborative research.	3	COMPBC	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
17	Jonas Kasmanas	jonas.kasmanas@ufz.de	From Gut to Data: Deep clustering the Human Microbiome for Fingerprints exploration and AutoML-Driven Bioindicator Discovery	Human gut microbiome profiles remain limited. Using dense deep clustering, we standardized 509,610 metagenome-assembled genomes (MAGs) to create microbiome fingerprints. Our resulting platform includes an AutoML-driven bioindicator detection, which can scan functional potential across countries and diseases. With 14,082 metagenomic runs (>20MM depth) from HumanMetagenomeDB, we recovered 302,781 MAG. We included 215,411 MAGs from published studies. After dereplication, we identified 6,794 species. All MAGs underwent gene annotation with ISfinder, NCBI-RefSeq, UniProt, and HMM databases. Next, we created a presence/absence gene profile for each MAG in adult gut samples (>18 years), yielding 426,648 profiles with 40,424 non-redundant genes. We implemented an autoencoder with dense layers and density-based clustering to capture metagenomic fingerprints. The embedded space revealed distinct microbiome profiles, showing significant differences between control and colorectal cancerous samples and separate libraries by country (PERMANOVA < 0.05). The MAGs clusters based on functional potential identified taxonomical groups diverging from their majority cluster, potentially signaling relevant functional shifts in specific strains. Our AutoML system uncovered geographical markers, with Lachnospira eligens, for instance, emerging as a promising bioindicator between Chinese and USA gut microbiomes. This study provided a standardized dataset, downloadable genome collection, and interactive webapp, expanding previously published resources for future microbiome research.	4	Department of Applied Microbial Ecology (AME)	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
18	Ricardo Siqueira da Silva	ricardo.siqueira@ufvjm.edu.br	ECOLOGICAL NICHE MODELING FOR BIOLOGICAL INVASION PREDICTIONS IN ENVIRONMENTAL CHANGE SCENARIOS FOR PRIORITIZED ABSENT QUARANTINE PESTS	The effects of climate change on exotic species may heighten threats to global agricultural production systems. Reported invasion costs have reached $1.288 trillion due to economic losses. Assessing the potential impact of climate change on invasive species is conducted using species distribution models under climate scenarios. Although species distribution models are widely used to evaluate the availability of favorable environments for establishing invasive exotic species, updates to these models for management and support of phytosanitary and conservation policy formulation with modern methods are lacking. Mapping current invasions and modeling suitable habitats of invasive species with improved models are important contributions to formulating strategies and policies to control the risk of invasive species. Species distribution models must be constantly updated in methodology and by including new occurrences to make species monitoring more informative. Using ecological niche models, we will select prioritized quarantine pest species in Brazil to determine the risks of species introduction. The results are useful for mitigating biological invasions' economic, social, and environmental impacts.	3	OESA	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs); Simulation I				
19	Jasmin Heilemann	jasmin.heilemann@ufz.de	Scenarios of the future irrigation water demand in Germany considering farmers' adaptive land use in a Multi-Agent System model	The predominantly rainfed agricultural sector in Germany is increasingly challenged by drought and heat impacts through climate change. Farmers need to adapt to these changing conditions, for instance by shifting to irrigated production systems. Expanding irrigation will increase the future irrigation water demand and may result in competition for water use and higher groundwater depletion. Simultaneously, farmers also adapt their land-use patterns, responding to changes in climatic and economic conditions. This land use adaptation affects the irrigation water demand. To project the future irrigation water demand in Germany considering adaptive land use, we present a spatial multi-agent system (MAS) model which simulates the rainfed and irrigated land use of eight major field crops using a positive mathematical programming (PMP) approach. Each agent captures the behavior of farmers in one of Germany's 401 districts and is individually calibrated to observed production conditions. To project the future land use and irrigation water demand, we use the SSP-RCP scenario framework, providing projections of future crop yields via a statistical crop model, driven by meteorological variables and soil moisture from the mesoscale Hydrologic Model (mHM), and crop price projections, based on the SSPs. Furthermore, the effect of different policies, such as increasing water tariffs and abtraction limits, are tested.	6	Economics (OEKON)	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs); Simulation I				
20	Michael Strauch	michael.strauch@ufz.de	SWAT+: A brief introduction into the Soil and Water Assessment Tool and how it is used in UFZ research	The Soil and Water Assessment Tool (SWAT) is widely used to assess the impact of agri-environmental practices (AEPs) on water and nutrient balance components at the catchment scale. In its new and fully revised version (SWAT+), the model allows users to customise flow paths and connect various landscape features. This includes more flexible routing of water, sediment and nutrients within and between different components of the catchment, providing a more realistic simulation of hydrological processes. Within the EU H2020 research project OPTAIN, we are using SWAT+ with continuous routing between all spatial units (e.g. individual agricultural fields and semi-natural landscape elements) to simulate surface and near-surface water flow and nutrient movement continuously across the landscape. This will allow the effects of AEPs (e.g. grassed waterways, buffer strips, conservation tillage, cover crops) to be studied in relation to their individual site-specific allocation within the catchment. Finally, we couple SWAT+ with the land use optimisation tool CoMOLA to identify optimal spatial plans of AEP implementation for multiple objective functions (e.g. minimising nutrient pollution, maximising water retention, maximising crop production, minimising AEP implementation costs).	1	CLE	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
21	Tobias Meisel	tobias.meisel@ufz.de	Workflow Management Software: Experiences from Research Projects	In complex computer-based scientific studies, it's crucial to ensure that processing steps are traceable, repeatable, and well-defined. Open-source workflow management software helps achieve this by organizing, automating, and executing tasks. These tools are flexible, scalable, and can integrate with existing systems, making them ideal for developing and using complex scientific workflows. In this presentation, we will provide an overview of solutions and experiences from various projects and discuss key factors to consider.	5	Environmental Informatics	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
22	Gokul Govind	gokul.govind@ufz.de	Enhancing Predictive Accuracy in Honeybee Colony Dynamics through Coupled mechanistic Simulations and Informed Machine Learning Models	To improve predictive accuracy in honeybee colony dynamics, we leverage a combination of BEEHAVE simulations, empirical modeling, and AI techniques within the BN-EXN project. By integrating extensive weather and crop diversity data from Germany, we create simulation datasets that capture key ecological dynamics, which then inform the training of machine learning (ML) and neural network (NN) models. This approach includes real-world and synthetic weather data, which are layered in NN architectures to reflect both temporal and spatial variation, achieving up to 92% accuracy in predicting annual honey yield compared to the simulations. A feature-importance analysis on non-spatial parameters in BEEHAVE supports a targeted modeling approach, enhancing the Digital Twin's predictive capabilities. Together, these methods can be utilized to transition the present BEEHAVE Digital Twin from a descriptive/partial-predictive to a fully functional predictive tool, creating valuable resources for beekeepers and farmers assisting the decision-making process. This work demonstrates how coupled informed Machine learning techniques incorporating domain knowledge and simulation methodologies can advance the scalability and functionalities of environmental digital twins, Individual-Based Models, and Agent-Based Models across multiple ecological applications.	5	Ecological Modelling	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
23	fan wu	fan.wu@ufz.de	/	/	1		AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
24	Jürgen Groeneveld	juergen.groeneveld@ufz.de	Digital Twins and mechanistic insect pollination models	Honey bees and other insect pollinators are vital to human well-being and are under multiple stresses. We need to be able to assess the viability and productivity of honey bee colonies in different landscapes and under different management and climate-change scenarios. Therefore, we have developed a prototype digital twin, HONEYBEE-pDT, based on the BEEHAVE model, which simulates foraging, population dynamics and Varroa mite infestation of a single honey bee colony. The main input data are land-cover maps and daily weather data. We have developed the pDT for simulating large areas and have tested it for the whole of Germany. We have also developed a web-based GUI that users can use to run the pDT for specific sites. Times series of hive weight data from hundreds of hives will be used for calibration and validation. For the calibration we plan to apply genetic algorithms. For the sensitivity analysis of the BEEHAVE model we have applied neural networks and the XGBoost algorithm to rank the importance of the model parameters. In ongoing work we aim to aggregate the mechanistic simulation model to have a more efficient tool at hand to compute scenarios on a national scale.	5	Ecological Modelling	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
25	David Manske	david.manske@ufz.de	Monitoring Renewable Energy Landscapes	Germany's energy system is transitioning to renewables, with renewable energy recently accounting for 22% of gross final energy consumption. The future system will be primarily electric, distributed, and highly interconnected. Renewable electricity generation is geographically uneven, concentrated in northern and eastern regions, while major consumption centers are located in the south and west. A successful energy transition requires contributions from all municipalities to establish a climate-friendly energy supply. Significant progress at federal and state levels has been achieved, but detailed data on local renewable energy use remains scarce. A new methodology has been developed to annually measure renewable energy generation and consumption at the municipal level. This approach enables detailed monitoring of renewable energy landscapes and the utilization of locally generated resources. Findings indicate notable rural-urban and north-south disparities in renewable energy production and consumption. Municipalities with population densities exceeding 100 inhabitants per km², comprising 88% of the population, cover only 22% of electricity needs through local renewable sources. Most renewable electricity is generated in sparsely populated areas. Addressing these imbalances requires enhancing transmission infrastructure and expanding urban renewable generation capacity. This methodology offers critical insights into decentralized energy contributions and supports the transition to a sustainable energy system.	6	BEN	Accounting Model (e.g., LCA); Simulation Model (e.g., agent-based models, econometrics and statistical analysis, ph				
26	Andrea Kaim	andrea.kaim@ufz.de	CoMOLA - Constrained Multi-objective Optimization of Land use Allocation	CoMOLA is a free Python tool to optimize the spatial allocation of land uses for multiple objectives and is based on the Non-dominated Sorting Genetic Algorithm II (NSGA-II). It includes functions for reading, encoding and writing land use maps as well as genome generation and repair mutation algorithms for considering constraints during the optimization procedure. It runs on Windows and Linux and allows for the integration of any model whose prediction (e.g. a value for an ecosystem service) is based on a land use raster map. The tool's output is a set of Pareto-optimal land use maps. Fields of application have been e.g. the spatial allocation of agricultural land uses such as different types of cropland and grassland maximizing yield, biodiversity and minimizing nutrient leaching, the optimal allocation of riparian reforestation efforts along rivers, exploring optimal strategies to retain water and nutrients in agricultural catchments. Other potential fields of application can be nature conservation, climate adaptation, restoration, urban planning, renewable energies or even other contexts outside of environmental research where spatial allocation of certain measures/uses are required. CoMOLA has also been used for teaching spatial multi-objective optimization at universities.	1	CLE	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs)				
27	Matthias Welker	matthias.m.welker@ufz.de	Regional biophysical boundaries and their application in Life-Cycle-Assessment studies – A review	Building on the concept of Planetary Boundaries, this research aims to review regional biophysical limits and discuss their applicability to Life-Cycle-Assessment-based Absolute Environmental Sustainability Assessment. While five out of the nine existing planetary boundaries are deemed to be relevant on the regional scale (water-use, biogeochemical flows, biosphere integrity, land-system change, and atmospheric aerosol loading), they only have been partly expressed with regional explicit quantifications. Simultaneously, Life-Cycle-Assessment is commonly applied to estimate the relative environmental impacts of production and consumption systems in similar impact categories, with deviating units of measurement, increasingly with a regional focus. This research aims to bring both together and review regionally expressible biophysical limits and assess their applicability to LCA-based assessment of absolute sustainability. The research hereby tries to bring together expertise from various research units with the overarching goal to enable the assessment of regionally explicit environmental impacts of scenarios and models of bio-based energy and material use.	6	BEN	Accounting Model (e.g., LCA)				
28	Ulisses Rocha	ulisses.rocha@ufz.de	Entering the twilight zone: expanding the boundaries of the protein annotation with language models	Annotating open reading frames (ORFs) remains challenging for alignment-based methods, particularly when dealing with remote homologs or highly divergent sequences. To address this, we introduce HiPfam, a novel framework that leverages advanced language models to improve functional annotation. HiPfam integrates hierarchical features from the SCOP and Pfam databases, facilitating enhanced protein family detection and the characterization of previously unannotated proteins. We compiled a dataset from the Collaborative Multi-domain Exploration of Terrestrial Metagenomes, encompassing over 16 million ORFs, dereplicated into 2.4 million orthologous groups. Of the 7 million representative sequences annotated with Pfam, KEGG, and COG, only 12.84% yielded annotations. To address this gap, we embedded 292,728 sequences using pre-trained language models and implemented a hierarchical multi-label classification approach based on SCOP-Pfam mappings of 1,375 SCOP families, employing the Local Classifier per Parent Node (LCPN) strategy. Models tested included ProSE-DLM, ProSE-MT, ProtBert-BFD, and ProtT5-XL-U50. Statistical analyses using the Friedman and Nemenyi tests confirmed that hierarchical classification improved smaller models like ProSE-DLM, achieving performance comparable to larger, complex models.	4	AME	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
29	Jörg Hackermüller	joerg.hackermueller@ufz.de	None	None	3	COMPBC	Accounting Model (e.g., LCA); AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
30	Ni Li	ni.li@vub.be	Wikimpacts 1.0: A new climate impact database using generative AI and Wikipedia	Storms, heat waves, wildfires, floods, and other climate-related hazards pose a significant threat to society and ecosystems. Understanding and quantifying the associated impacts is thus a critical scientific and societal challenge. While impact databases are needed for linking climate hazards to socio-economic impacts, publicly available impact data remain scarce. Beyond existing open databases such as EM-DAT, robust data on the impacts of climate extremes can also be found in textual documents, such as newspapers, reports and Wikipedia articles. Here we present Wikimpacts 1.0, a new climate impact database based on English Wikipedia and the large language model (LLM) GPT4o. Built using a pipeline including documentation selection, key information extraction, post-processing, evaluation, and consolidation, our approach allows for the generation of a comprehensive impact database. Significantly, our dataset encompasses 3366 climate extreme events spanning the years 1034 to 2024. Regarding impact information, it includes 18,233 records at the national level and 58,211 records at the sub-national level. The sub-national level records provide more impact information on Extratropical Storms/Cyclones, Tropical Storms/Cyclones, Tornadoes, and Wildfires compared to the benchmark database EM-DAT in the same period (1900-2024). Furthermore, there is less missing impact data for the documented events in countries such as the United States, Canada, and Australia. Our research highlights the opportunities of natural language processing to collect data on climate impacts, which can complement existing open impact datasets to provide more robust information on the impacts of weather and climate events.	1	Urban and Environmental Sociology	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
31	Mansi Nagpal	mansi.nagpal@ufz.de	Measuring crop acreage adaptation to changing yields and prices in Germany using DroughtMAS	The pressures of frequent extreme weather like droughts pose a continued challenge to maintaining the sustainable development of agricultural systems. The expected increase in human-induced climate change will exacerbate this challenge, making farmer adaptation and large-scale changes in production systems and land use inevitable. A key coping strategy farmers adopt includes changes in the choice of crops and acreage to adapt to climate change and minimize its negative effects. However, these adaptations are rarely quantified or demonstrated at the country scale. Our goal is to measure the degree to which crop acreages adapt to changing prices and yields. To achieve this, we introduce a metric that quantifies cropping adaptation by measuring the number of hectares differing from an economically optimal cropping pattern. This economically optimal cropping pattern is operationalized using the econometric mathematical programming approach within the spatial multi-agent system model DroughtMAS framework. The model simulates profit-maximizing cropping allocation under given economic production conditions while calibrating to the observed cropping behavior. Individually parameterized land-user agents and agent-specific calibration coefficients capture the biophysical and agro-economic heterogeneity at the district level. By applying our concept to Germany, we find (1) persistent crop acreage changes by 7850 hectares on average per district between 1999 and 2020. (2) Average deviation of 4715 hectares per district from an economically optimal cropping pattern between 1999 and 2020. (3) crop acreages in any year are more adapted to current prices and yields than in earlier years and (4) cropping adaptedness in drought land use significantly lower than in non-drought land-use. The results highlight the dynamics of cropping adaptation over time and improve the understanding of the effects of climate change on cropping areas. A significant difference in cropping adaptation for land use under drought and non-drought conditions sheds light on bigger adaptation challenges under extreme events. These results provide empirical insights for shaping future policies to support the adaptation of agriculture to weather extremes and climate change and promote sustainable management of land and water resources.	6	Economics	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs); Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
32	Husain Najafi	husain.najafi@ufz.de	Advancing High-Resolution Flood and Drought Forecasting: Insights from UFZ's Hydrological Models and Data Systems	Hydrological and impact forecasting models at UFZ are advancing the prediction of floods and droughts with high-resolution accuracy. To date, over 100 TB of data has been generated to evaluate and enhance the predictability of these extreme events. This contribution outlines the complete forecasting chain employed, highlighting the integration of data, model components, and computational workflows. Furthermore, it addresses the associated uncertainties, offering insights into their implications for model performance and decision-making under extreme climate scenarios. The methods demonstrated here showcase their scalability and potential adaptability to other environmental and research domains.	5	CHS	Optimization Model (e.g., linear programming, mixed-integer programming, multi-level models, IAMs); Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				
33	Lennart Schmidt	lennart.schmidt@ufz.de	Preprocessing and model building strategies to tackle common obstacles in machine learning applications for urban waste water systems	Combined sewer systems are influenced by increasingly frequent stormwater events due to climate change. Thus, automated tools for system monitoring and control are essential to minimize emissions into surface waters. Machine learning algorithms are commonly showcased as promising tools but their adoption in practice is often hindered by 1) inconsistent data management of in-situ measurements (data gaps, inconsistent measurement frequencies and measurement set-ups) and 2) system dynamics that are challenging to represent in statistical models (skewed data distributions, heterogeneous variable dynamics). We modeled the sewer system of a German city in an autoencoder model and investigated both preprocessing and model building strategies to overcome the aforementioned challenges. The best results were achieved using a combination of missing value masking, careful expert-informed interpolation and separate models for each variable group that predict increments over time instead of raw data values. By extending our findings with data management recommendations for operators, we provide guidance on circumventing little-reported but critical obstacles to the scalable application of data-driven solutions in the urban water sector.	5	MET	AI and Data-Driven Model (e.g., CNN, DQN, RL, LLM)				
34	Karin Frank	karin.frank@ufz.de	AI and Simulation Models: Some thoughts from the Department OESA	This talk will highlight running and planned activities at the Department of Ecological Modelling (OESA) in the newly emerging research field of 'Linking AI/ML with Simulation Models'. OESA contributes to the UFZ's research on the multifunctionality of landscapes under transition. It focusses on the development of simulation models for predicting and causally understanding how ecological and social-ecological systems respond to changing environmental conditions. In this domain, individual-based (IBM) resp. agent-based (ABM) modeling approaches are essential. A recent literature, however, underpins that there are various challenges in the context of IBMs/ABMs which can substantially benefit from an appropriate linkage with AI/ML approaches. In the talk, we will, first, stress the importance of Informed ML aiming at the integration of domain knowledge and causalities in the ML pipelines. We will, second, present OESA's plan to adjust InformedML to IBM/ABM simulation models by (i) using their main components (causal diagram, model rules, patterns for validation) as assets for the knowledge integration, (ii) simulation output as synthetic training data, and (iii) published insights derived from IBMs/ABMs for testing the performance of the adjusted InformedML-algorithms. Third, we will invite for a planned Journal Club on "AI/ML and simulation models" for review and/or perspective papers. Hint: this talk should come after the talk by Gokul Govind on the development of an emulator.	5	Department of Ecological Modelling (OESA)	Simulation Model (e.g., agent-based models, econometrics and statistical analysis, physics-based models, IAMs)				